

Step 01 — Project scaffold (manual test guide)

Goal: Confirm PERN scaffold works: Express API, React + Tailwind UI, Prisma wired to PostgreSQL.

Maps to: Phase 1 — Foundation

Do not start Step 02 until every required item passes.

Artifacts:

- Markdown: docs/manual-test-guides/step-01-project-scaffold.md
- PDF: docs/manual-test-guides/step-01-project-scaffold.pdf

Prerequisites

- ☐ Node.js 20+ installed (`node -v`)
- ☐ PostgreSQL running locally
- ☐ Database created (example): `fms_dev`
- ☐ `backend/.env` has a valid `DATABASE_URL` (copy from `.env.example` if needed)

Example `DATABASE_URL` :

```
postgresql://USER:PASSWORD@localhost:5432/fms_dev?schema=public
```

Setup commands

From `finance-management/backend` :

```
npm install
npx prisma generate
npx prisma migrate dev --name step01_health
npm run dev
```

From `finance-management/frontend` (new terminal):

```
npm install
npm run dev
```

Checklist

A. Backend boots

#	Test	Expected	Pass?
A1	Run <code>npm run dev</code> in backend/	Console shows FMS API listening on <code>http://localhost:4000</code>	[]
A2	Open <code>http://localhost:4000/</code>	JSON with <code>ok: true</code> and message about FMS API	[]
A3	Folder check	You can find <code>routes/</code> , <code>controllers/</code> , <code>services/</code> , <code>middleware/</code> , <code>validators/</code> , <code>config/</code> under <code>backend/src/</code>	[]

B. Database connectivity

#	Test	Expected	Pass?
B1	<code>npx prisma migrate dev</code> succeeds	Migration applied without error	[]
B2	GET <code>http://localhost:4000/api/health</code>	HTTP 200, <code>"ok": true, database.connected: true</code>	[]
B3	Stop Postgres (or break <code>DATABASE_URL</code>) and hit <code>/api/health</code> again	HTTP 503 (or ok false), <code>database.connected: false</code> with an error message	[]
B4	Restore Postgres / correct URL	Health returns connected again	[]

C. Frontend + Tailwind

#	Test	Expected	Pass?
C1	Run <code>npm run dev</code> in frontend/	App at <code>http://localhost:5173</code>	[]
C2	Page loads	Heading "Step 01 scaffold is running" and teal accent styling (Tailwind)	[]
C3	With API running	Status section shows API OK and DB connected	[]
C4	Stop API, refresh page	UI shows an error about failing to reach API	[]
C5	Reusable UI folder	<code>frontend/src/components/ui/Button.tsx</code> exists	[]

D. Coding rules smoke check

#	Test	Expected	Pass?
D1	Controllers call services	<code>healthController</code> uses <code>healthService</code> (not raw Prisma in the route)	[]
D2	Arrow-style handlers	Controllers/services use arrow function exports	[]

Pass criteria for Step 01

- ☐ A1–A3 pass
- ☐ B1–B2 and B4 pass (B3 optional but recommended)
- ☐ C1–C5 pass
- ☐ D1–D2 pass

When all required items pass: mark related Notion Phase 1 scaffold tasks Done, then start **Step 02 — Auth**.

Notes / failures

Item	What failed	Fix applied

Optional PDF

Print or export this Markdown to PDF if you want a paper checklist:

<docs/manual-test-guides/step-01-project-scaffold.pdf>